CIGNUS Application: Security and Architecture Documentation

Prepared in response to KRC's IT team documentation request

Prepared by	Fute Services IT Team
Prepared for	KRC IT Team
Project	CIGNUS, real estate sales presentation application
Attention	Ms. Soma Ratheesh
Classification	Confidential, Client Use Only

1. Overview

CIGNUS is a sales presentation application built for a real estate development. It is built for two device targets: Android tablets, and Windows desktop machines. A sales representative uses it to walk a prospective buyer through the property, including walkthrough video, drone footage, a 360 degree virtual tour of key locations, floor plans, and PDF brochures.

The application does not require an internet connection while it is being used. This was a deliberate design decision rather than an oversight: property viewings and sales meetings often take place in sample flats, basements, or site offices where a reliable connection cannot be guaranteed. Every video, image, and document the app needs is packaged into the installer at build time. The trade-off is a larger install size, and the fact that any content update requires a new build and a fresh install rather than a server-side push.

The Windows build is produced using Electron, wrapped around the same underlying application code used for the Android tablet build, so both targets are maintained from a single codebase.

This document, the CIGNUS source code, and all related project material are confidential and remain the property of Fute Services and the CIGNUS client. It has been prepared solely for KRC's IT team as part of vendor documentation review, and should not be copied or forwarded outside the reviewing team without written consent from Fute Services.

2. Data Classification

Application content (video, VR imagery, floor plans, PDF brochures)	Public marketing material, compiled into the application at build time. It carries no sensitivity beyond ordinary marketing confidentiality, and is not user data.
End-user data	None is collected. There is no login, no account system, no form submission, and no telemetry anywhere in the codebase.
Source code and internal project documentation	Confidential and restricted to the Fute Services and CIGNUS project teams. This is a separate classification from the "no end-user data" finding above, since the source code carries its own sensitivity independent of what the finished application handles at runtime.

3. Network Architecture PROVIDED

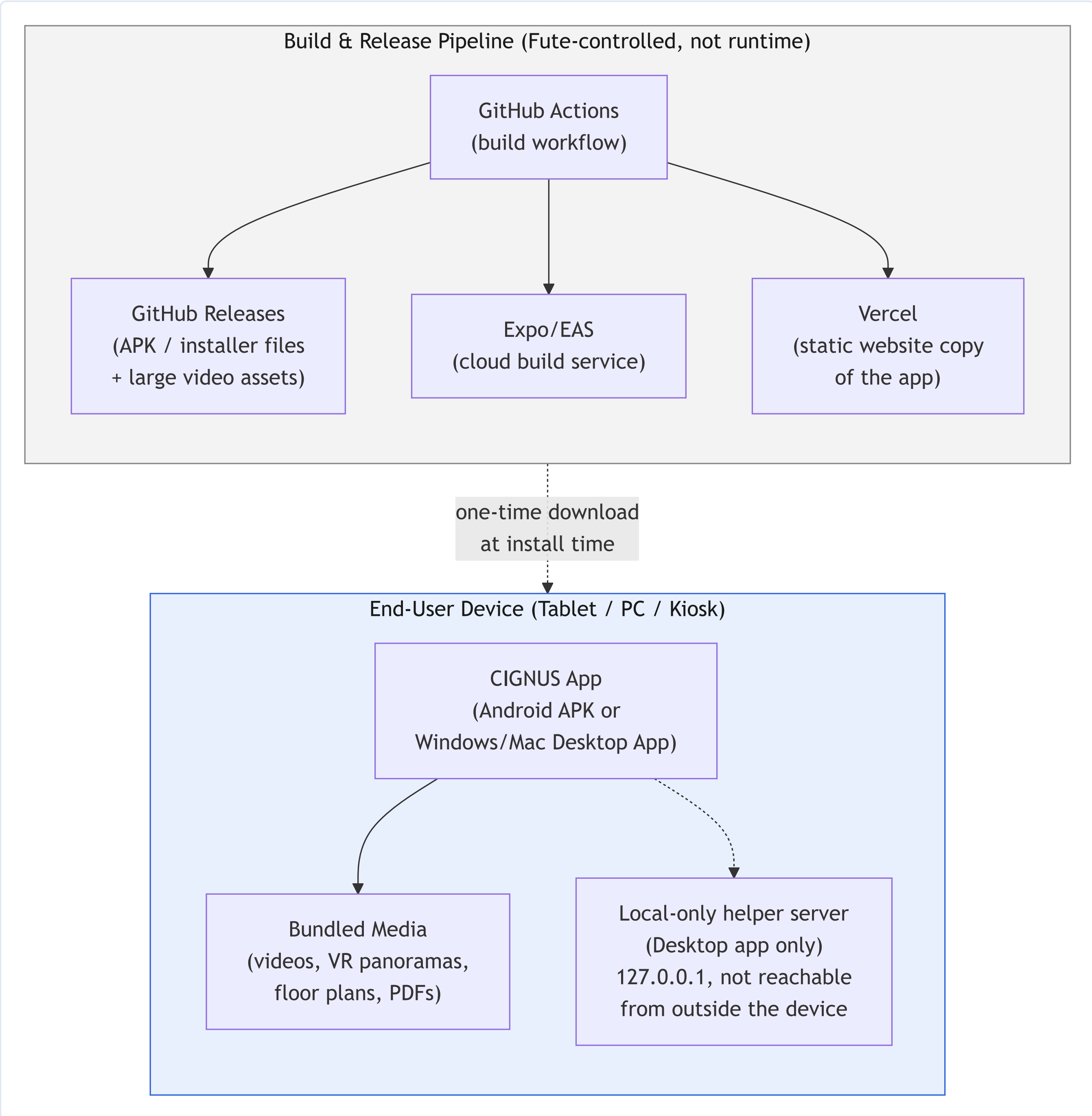

Figure 1: The application runs entirely on the user's device. Network activity is confined to the build and release pipeline, which is controlled by Fute Services.

Once installed, the application has no live backend to talk to. There is no company server involved while a sales representative is using it on a tablet or a laptop. The only place a network connection is actually used is behind the scenes, during the build and release process: GitHub Actions builds the Android package and the Windows installer, publishes them to GitHub Releases, and the Expo web export is deployed as a static site through Vercel.

On the Windows build, Electron starts a small local server bound to 127.0.0.1 so the packaged web assets can be served the same way they would be in a browser. That server never listens on any network interface other than the loopback address, so it cannot be reached from another device on the same network, let alone from the internet.

4. Data Flow PROVIDED

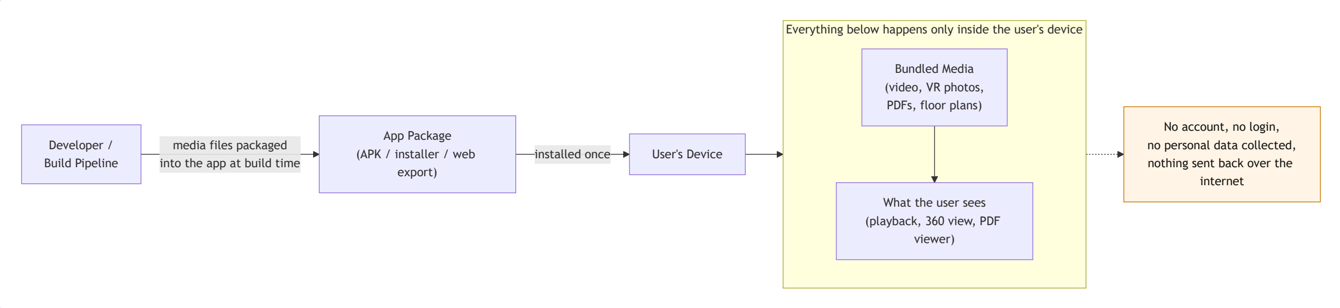

Figure 2: Media is packaged into the application once, at build time, and is only ever displayed on the device. Nothing flows back out.

All video, VR imagery, and documents are packaged into the application when it is built. From that point forward, the flow of information is one directional: from the installed package to the user's screen. Nothing about the user, the device, or how the application is used is collected, stored, or transmitted anywhere.

5. Mobile Application Testing COMPLETED

Testing was carried out by the Fute Services IT team, working directly against the codebase and build configuration rather than against an installed device. This distinction matters for the purpose of this review: a code and configuration review catches structural issues such as unsafe permissions, embedded credentials, or unsafe content loading. It does not replace a dynamic assessment run against a live, installed build under real device conditions.

What was reviewed:

- **Dependency audit.** The application's dependency tree was checked for known vulnerabilities. The issues found are limited to development and build tooling packages, and do not affect the code shipped inside the final application.
- **Device permission review.** The Android build requests Internet access, read and write access to external storage, vibration, and the ability to display over other apps, used respectively for content loading, file handling, haptic feedback, and video playback. It does not request access to contacts, camera, microphone, location, or SMS.
- **Content loading review.** The VR panorama images shown inside the application are loaded exclusively from files bundled inside the application package. None of them are fetched from an external or remote source at runtime.

6. Source Code Review COMPLETED

A source code review was carried out by the Fute Services IT team, covering the following areas:

- **Credential scan.** The full codebase was scanned for hardcoded API keys, passwords, tokens, and private keys. None were found.
- **Desktop application security configuration.** The Windows build, built on Electron, was reviewed against current Electron security practice: the renderer process runs in an isolated context with no direct access to Node.js APIs, and the browser window itself runs sandboxed.
- **External navigation handling.** The desktop application blocks any attempt, from within the application, to navigate to or open a web address outside its own local server.

7. API Security Testing NOT APPLICABLE

This item does not apply to CIGNUS. The application has no backend, and does not call, expose, or connect to any API. There is no server-side component anywhere in the project. Since API testing is only relevant where an API exists, none was carried out.

8. Project Coordination

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